

Scientific Programming Lab - Unit 2

1. Write a program to demonstrate working with strings in python

Question: Write a program to demonstrate working with strings in python.

Aim: To perform operations on strings like concatenation, slicing, and length.

Objective: To understand basic string operations in Python.

Program:

```
s1 = input("Enter first string: ")
s2 = input("Enter second string: ")
print("Concatenation:", s1 + s2)
print("Length of first string:", len(s1))
print("First three characters of second string:", s2[:3])
print("Reversed first string:", s1[::-1])
```

Result/Output:

Enter first string: Hello

Enter second string: World

Concatenation: HelloWorld

Length of first string: 5

First three characters of second string: Wor

Reversed first string: olleH

Result Analysis: The program gives the desired output according to the objective successfully.

2. Write a program to create, append, and remove lists in python

Question: Write a program to create, append, and remove lists in python.

Aim: To create a list and demonstrate append and remove functions.

Objective: To understand list manipulation in Python.

Program:

```
my_list = []
my_list.append(10)
my_list.append(20)
my_list.append(30)
print("List after appending:", my_list)
my_list.remove(20)
```

```
print("List after removing 20:", my_list)
```

Result/Output:

List after appending: [10, 20, 30]

List after removing 20: [10, 30]

Result Analysis: The program gives the desired output according to the objective successfully.

3. Write a program to demonstrate working with tuples in python

Question: Write a program to demonstrate working with tuples in python.

Aim: To define and access elements of a tuple.

Objective: To understand tuple properties and usage.

Program:

```
t = (1, 2, 3, 4, 5)
print("Tuple:", t)
print("First element:", t[0])
print("Last element:", t[-1])
print("Sliced tuple (2nd to 4th):", t[1:4])
```

Result/Output:

Tuple: (1, 2, 3, 4, 5)

First element: 1

Last element: 5

Sliced tuple (2nd to 4th): (2, 3, 4)

Result Analysis: The program gives the desired output according to the objective successfully.

4. Write a program to demonstrate working with dictionaries in python

Question: Write a program to demonstrate working with dictionaries in python.

Aim: To create a dictionary and perform basic operations.

Objective: To understand dictionary creation, accessing, and updating.

Program:

```
d = {"name": "Adil", "age": 20, "branch": "CSE"}
print("Dictionary:", d)
print("Name:", d["name"])
d["age"] = 21
print("Updated Dictionary:", d)
```

```
d["city"] = "Hyderabad"
print("Extended Dictionary:", d)
```

Result/Output:

Dictionary: {'name': 'Adil', 'age': 20, 'branch': 'CSE'}

Name: Adil

Updated Dictionary: {'name': 'Adil', 'age': 21, 'branch': 'CSE'}

Extended Dictionary: {'name': 'Adil', 'age': 21, 'branch': 'CSE', 'city': 'Hyderabad'}

Result Analysis: The program gives the desired output according to the objective successfully.

5. Stack implementation in python

Question: Stack implementation in python.

Aim: To implement stack operations using list.

Objective: To demonstrate LIFO (Last In First Out) behavior using list.

Program:

```
stack = []
stack.append('A')
stack.append('B')
stack.append('C')
print("Stack after pushing:", stack)
stack.pop()
print("Stack after popping:", stack)
```

Result/Output:

Stack after pushing: ['A', 'B', 'C']

Stack after popping: ['A', 'B']

Result Analysis: The program gives the desired output according to the objective successfully.

6. Queue implementation in python

Question: Queue implementation in python.

Aim: To implement queue operations using list.

Objective: To demonstrate FIFO (First In First Out) behavior using list.

Program:

```
queue = []
queue.append('X')
```

```
queue.append('Y')
queue.append('Z')
print("Queue after enqueue:", queue)
queue.pop(0)
print("Queue after dequeue:", queue)
```

Result/Output:

Queue after enqueue: ['X', 'Y', 'Z']

Queue after dequeue: ['Y', 'Z']

Result Analysis: The program gives the desired output according to the objective successfully.

7. Linear and Binary search programs in python

Question: Linear and Binary search programs in python.

Aim: To implement linear and binary search algorithms.

Objective: To understand search operations on lists.

Program (Linear Search):

```
def linear_search(arr, target):
    for i in range(len(arr)):
        if arr[i] == target:
            return i
    return -1

arr = [10, 20, 30, 40, 50]
target = 30
print("Linear Search Result:", linear_search(arr, target))
```

Program (Binary Search):

```
def binary_search(arr, target):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
```

```
        low = mid + 1
    else:
        high = mid - 1
    return -1
```

```
arr = [10, 20, 30, 40, 50]
target = 40
print("Binary Search Result:", binary_search(arr, target))
```

Result/Output:

Linear Search Result: 2

Binary Search Result: 3

Result Analysis: The program gives the desired output according to the objective successfully.

8. Selection and Bubble sort programs in python

Question: Selection and Bubble sort programs in python.

Aim: To sort elements using selection sort and bubble sort algorithms.

Objective: To understand basic sorting algorithms in Python.

Program (Selection Sort):

```
def selection_sort(arr):
    for i in range(len(arr)):
        min_idx = i
        for j in range(i+1, len(arr)):
            if arr[j] < arr[min_idx]:
                min_idx = j
        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr
```

```
arr = [64, 25, 12, 22, 11]
print("Selection Sorted Array:", selection_sort(arr))
```

Program (Bubble Sort):

```
def bubble_sort(arr):
    n = len(arr)
```

```
for i in range(n):  
    for j in range(0, n-i-1):  
        if arr[j] > arr[j+1]:  
            arr[j], arr[j+1] = arr[j+1], arr[j]  
    return arr
```

```
arr = [64, 34, 25, 12, 22, 11, 90]  
print("Bubble Sorted Array:", bubble_sort(arr))
```

Result/Output:

Selection Sorted Array: [11, 12, 22, 25, 64]

Bubble Sorted Array: [11, 12, 22, 25, 34, 64, 90]

Result Analysis: The program gives the desired output according to the objective successfully.